Chapter 1. Finite Groups

Part I - Representations and Schur's Lemma

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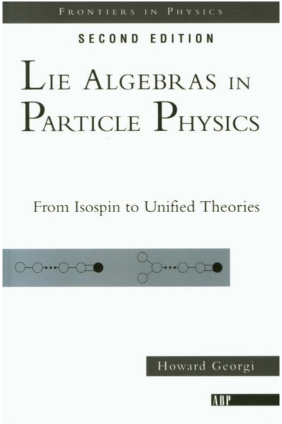
Outline

Introduction

"*Oh these are old fogeys; in five years every student will learn group theory as a matter of course.*"

John von Neumann

- **Textbook :** Georgi, *Lie Algebras in Particle Physics*



Outline

Group

Group

A **group** $(G, \cdot)$ is a nonempty set G with a binary operation $\cdot : G \times G \to G$ such that

1. $\forall a, b \in G$, $ab \equiv a \cdot b \in G$ (Closure)
2. $a(bc) = (ab)c$ (Associativity)
3. $\exists e \in G$ s.t. $\forall a \in G, ae = ea = a$ (Identity)
4. $\forall a \in G$, $\exists a^{-1} \in G$ s.t. $aa^{-1} = a^{-1}a = e$ (Inverse)

- A group is *commutative* or *Abelian* if $\forall a, b \in G$, $ab = ba$
- Given $a \in G$, e and a^{-1} are unique
- We usually omit $\cdot$ and write G solely for groups
- We will mostly think of groups as collections of **symmetry transformations**

Subgroup

Subgroup

A subset H of a group G is called a **subgroup** if H is a group under operation of G and written $H \leq G$.

- By definition, $\{e\}$ and G are subgroups of G, called **trivial subgroups**

We may use a subgroup H to 'divide up' the elements of the group G into subsets called **cosets**, and form a new group called **factor group** G/H

Cosets

For a $H \leq G$ and any $g \in G$, let

$$gH = \{gh | h \in H\}, \quad \text{and} \quad Hg = \{hg | h \in H\}$$

called respectively a **left coset** and a **right coset** of H in G.

Subgroup - Example

Integer group $(\mathbb{Z}, +)$ has $n\mathbb{Z} = \{na | a \in \mathbb{Z}\}$ as subgroups for all integral n

$$0 + n\mathbb{Z}, \quad 1 + n\mathbb{Z}, \quad \cdots (n-1) + n\mathbb{Z}$$

are *left* cosets of $n\mathbb{Z}$ in $\mathbb{Z}$

$$\begin{aligned}
[0] &\equiv 0 + n\mathbb{Z} = \{\cdots, -n, 0, n, 2n, 3n, 4n, \cdots\} \\
[1] &\equiv 1 + n\mathbb{Z} = \{\cdots, 1-n, 1, 1+n, 1+2n, 1+3n, 1+4n \cdots\} \\
&\vdots
\end{aligned}$$

So *any* integer a must be a member of exactly one of the cosets $[0], [1], \cdots, [n-1]$ and actually

$$\mathbb{Z} = [0] \sqcup [1] \sqcup [2] \sqcup \cdots \sqcup [n-1]$$

Subgroup - Example

Define 'addition' of cosets as follows

$$[a] + [b] = [a + b], \quad 0 \leq a, b \leq n - 1$$

Surprisingly, the set $\{[0], [1], \cdots, [n-1]\}$ forms a group under 'addition' defined above
This is precisely '**addition modulo** n'

$$[2] + [3] = [5] = [1] \quad \text{in } \mathbb{Z}/4\mathbb{Z} \iff 2 + 3 = 5 = 1 \quad \text{mod } 4$$

Our new group is called a **factor group** and denoted $\mathbb{Z}/n\mathbb{Z}$ (e.g. $\mathbb{Z}/2\mathbb{Z} = \{0, 1\} \cong \mathbb{Z}_2$)

Question : *Is this construction valid for all groups*?

Subgroup - Example

Our construction relied on the well-definedness of our special 'addition' of cosets $[a]$
$\rightarrow$ Is it well-defined? i.e. If $[a] = [a'], [b] = [b']$ then

$$[a] + [b] = [a'] + [b']?$$

Suppose our subgroup N is special, satisfying :

$$gNg^{-1} = N, \quad \forall g \in G$$

Then our 'operation' for cosets takes the form

$$(aN)(bN) = abb^{-1}NbN = (ab)N$$

So the condition $gNg^{-1} = N$ was **essential** for our construction

Normal Subgroup and Factor Group

Normal Subgroup

A subgroup N of a group G is called a **normal subgroup** if

$$gN = Ng \quad \forall g \in G. \tag{1}$$

We write $N \trianglelefteq G$ for normal subgroups.

Factor Group

Let $N \trianglelefteq G$ be a normal subgroup. The set of cosets $G/N = \{gN | g \in G\}$ forms a group under an operation

$$(g_1 N)(g_2 N) = (g_1 g_2) N.$$

The group G/N is called a **quotient group** or **factor group**.

Representations

Representation

A representation of a group G is a mapping $D : G \to GL(V)$[a] with the following properties.

1. $D(e) = \mathbf{1}$,
2. $D(g_1)D(g_2) = D(g_1 g_2)$.

The **dimension** of a representation is the dimension of the vector space V. Furthermore, a representation is **faithful** if D is injective.

[a] $GL(V)$ denotes the set of all invertible linear transformations $D(g) : V \to V$.

- Roughly, a representation gives a 'matrix realization' of the symmetry transformations
- Unless otherwise stated, all representations are assumed to be **finite-dimensional** and **complex**

Representations - Examples

- The **trivial representation** $D(g) = \mathbf{1}, \quad \forall g \in G$ is a representation, which is *not* faithful
- The following is a complex 1-dim rep of $\mathbb{Z}_n = \left\{e, a, a^2, \cdots, a^{n-1}\right\}$:

$$D(e) = 1, \quad D(a^m) = e^{(2\pi i)m/n}$$

- Let $\mathbf{R}(\theta) \in SO(2)$ denote a planar rotation by θ

$$D(\mathbf{R}(\theta)) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

  is a **real** 2-dim rep of the given group

Regular Representation

Let G be a finite group, and take the group elements to form an orthonormal basis for a vector space : $\{|e\rangle, |g_1\rangle, \cdots\}$, and define

$$D(g_1)|g_2\rangle = |g_1 g_2\rangle$$

Then

$$D(e)|g\rangle = |eg\rangle = |g\rangle = \mathbf{1}|g\rangle\,,$$
$$D(g_1 g_2)|g\rangle = |g_1 g_2 g\rangle = D(g_1)D(g_2)|g\rangle$$

Hence this is a representation, called the **regular representation**

Regular Representation

The matrix elements can be computed as follows :

$$[D(g)]_{ij} = \langle e_i | D(g) | e_j \rangle$$

Also we can compute $[D(g_1 g_2)]_{ij}$ by insertion of a complete set of intermediate states

$$\begin{aligned} [D(g_1 g_2)]_{ij} &= \sum_k \langle e_i | D(g_1) | e_k \rangle \langle e_k | D(g_2) | e_j \rangle \\ &= \sum_k [D(g_1)]_{ik} [D(g_2)]_{kj} \end{aligned}$$

This procedure is completely general for any finite group

Invariant Subspace

Invariant Subspace

Let $A : V \to V$ be a linear map. A subspace $W \leq V$ is called an (A-) **invariant subspace** if $\forall w \in W$, $Aw \in W$.

- Consider a projection operator P, projecting onto W

$$Pv \in W \quad \forall v \in V \implies D(g)Pv \in W \quad \forall g \in G$$

Since W is an invariant subspace. Since $Pw = w \quad \forall w \in W$, we have this operator equation

$$PD(g)P = D(g)P \quad \forall g \in G. \tag{2}$$

This condition is equivalent to the definition of invariant subspace

Irreducible Representation

Irreducible Representation, Completely Reducible Representation

A representation D is **reducible** if it has a nontrivial invariant subspace, i.e. there exists a proper subspace W such that

$$\forall w \in W, D(g)w \in W \quad \forall g \in G.$$

A representation is **irreducible** if it is *not* reducible.
A representation is **completely reducible** if it is equivalent(similar) to direct sum[a] of irreducible (sub)representations, i.e.

$$D \cong \bigoplus_i D_i. \quad (i : \text{irrep label}) \tag{3}$$

[a]For a detailed discussion, see appendix [45].

Irreducible Representation - Block-Diagonal Form

- Completely reducible representations are equivalent to a representation whose matrix elements have the **block-diagonal form** :

$$\begin{bmatrix} \boxed{D_1(g)} & & & \\ & \boxed{D_2(g)} & & \\ & & \boxed{D_3(g)} & \\ & & & \ddots \end{bmatrix} \begin{bmatrix} \boxed{W_1} \\ \boxed{W_2} \\ \boxed{W_3} \\ \vdots \end{bmatrix} = \begin{bmatrix} \boxed{D_1(g)\,W_1} \\ \boxed{D_2(g)\,W_2} \\ \boxed{D_3(g)\,W_3} \\ \vdots \end{bmatrix}.$$

Figure 1: The chosen invariant subspace W_2 does not mix with other subspaces under $D(g)$.

- **Warning** : The same irrep can appear more than once!
- All 1-dimensional reps are irreps. (why?)

Useful Theorems

Useful Theorems

1. Every representation of a finite group is equivalent to a unitary representation.
2. Every representation of a finite group is completely reducible.

proof. See [42] for proof of (1). We show $(1) \implies (2)$. WLOG assume unitary rep D.

Case 1. D is an irrep. Then we are done.

Case 2. D is reducible. Then there exists an invariant subspace W with associated projector P. We show that W's complement, $W^{\perp}$ is also an invariant subspace.

$$
\begin{aligned}
PD(g)P = D(g)P &\implies PD^{\dagger}(g)P = PD(g)^{\dagger} \quad \forall g \in G \\
&\iff PD(g^{-1})P = PD(g^{-1}) \quad \forall g \in G \\
&\iff PD(g)P = PD(g) \quad \forall g \in G
\end{aligned}
$$

Useful Theorems

The last line implies that $W^\perp$ is also an invariant subspace, since associated projector $(1-P)$ satisfies

$$(1-P)D(g)(1-P) = D(g)(1-P).$$

Therefore $V = W \oplus W^\perp$. We can repeat this process until V becomes completely reducible. This procedure must terminate, since the dimension of the remaining reducible subspace is strictly decreasing. $\square$

$$V = \begin{pmatrix} W & 0 \\ 0 & W^\perp \end{pmatrix} \to \begin{pmatrix} \begin{pmatrix} W' & 0 \\ 0 & W'^\perp \end{pmatrix} & 0 \\ 0 & W^\perp \end{pmatrix} \to \cdots$$

Figure 2: A schematic diagram of the proof.

Outline

Schur's Lemma

Schur's Lemma

Let $D_1(g) : G \to GL(V)$ and $D_2(g) : G \to GL(V')$ be complex irreps[a] of G. If $A \in \mathcal{L}(V, V')$ is such that

$$AD_1(g) = D_2(g)A \quad \forall g \in G, \tag{4}$$

then either A is an isomorphism or $A = 0$.

[a]This version of Schur's lemma equally applies for infinite-dimensional reps.

proof. Suppose $|a\rangle \in \ker A$. Then

$$AD_1(g)\,|a\rangle = D_2(g)A\,|a\rangle = 0 \implies D_1(g)\,|a\rangle \in \ker A \quad \forall g \in G.$$

Therefore $\ker A \leq V$ is invariant under $D_1(g)$. Irreducibility of $D_1(g)$ implies that $\ker A = V$ or 0.

Schur's Lemma

(Case 1) $\ker A = V$. In this case A is the zero map.

(Case 2) $\ker A = 0$. Then A is injective. Now we show that A is surjective. Let $|b\rangle \in \mathrm{Im}\, A$. Then $|b\rangle = A\,|x\rangle$ for some $|x\rangle \in V$ and

$$D_2(g)\,|b\rangle = D_2(g)A\,|x\rangle = AD_1(g)\,|x\rangle \implies D_2(g)\,|b\rangle \in \mathrm{Im}\, A.$$

Therefore $\mathrm{Im}\, A$ is invariant under $D_2(g)$. Irreducibility of $D_2(g)$ implies that $\mathrm{Im}\, A = V'$ or 0. Hence A is surjective. $\square$

Schur's lemma can be restated as follows :

Schur's Lemma (1/2)

Let $D_1(g)$ and $D_2(g)$ be inequivalent irreps of G. If $AD_1(g) = D_2(g)A \quad \forall g \in G$, $A = 0$.

Schur's Lemma

Schur's Lemma (2/2)

Let $D : G \to GL(V)$ be a complex irrep of G. If $AD(g) = D(g)A \quad \forall g \in G$, then $A = \lambda \mathbf{1}$.

proof.

(Case 1) $A = 0$. Let $\lambda = 0$.

(Case 2) A is an isomorphism. Since A is invertible, $\det A \neq 0$ and A must have nonzero eigenvalue λ with at least one eigenvector. But by assumption the operator $A - \lambda\mathbf{1}$ commutes with all $D(g)$. Since $\det(A - \lambda\mathbf{1}) = 0$, it is not an isomorphism and it must be the zero operator. $\square$

Consequences of Schur's Lemma

1. Rewrite Schur's lemma (2/2) as

$$A^{-1}D(g)A = D(g) \quad \forall g \in G \implies A \propto I$$

   This implies that once D is fixed, form of basis states are **unique** (up to common phase factor)

2. Suppose we have chosen a basis $|a, j, \mathbf{x}\rangle$, making D block-diagonal :

$$D = \begin{bmatrix} D_1(g) & & \\ & D_2(g) & \\ & & \ddots \end{bmatrix}$$

   where a is an irrep label, j is the state label within $D_a(g)$, but what is $\mathbf{x}$?

Consequences of Schur's Lemma

$$D(g) = \begin{bmatrix} \underbrace{\boxed{D_1(g)} & & & \\ & \boxed{D_1(g)} & & \\ & & \ddots & \\ & & & \boxed{D_1(g)}}_{x=1,\ldots,m_1} & & \\ & & & & \boxed{D_2(g)} & \\ & & & & & \ddots \end{bmatrix} \tag{5}$$

- **Multiplicity** must be considered, and $\mathbf{x}$ is the multiplicity label
- From now on, we will use this canonical block form for all reps

Consequences of Schur's Lemma

In this basis, matrix elements of $D(g)$ are

$$\langle a, j, x|\, D(g)\, |b, k, y\rangle = \delta_{ab}\delta_{xy}[D_a(g)]_{jk}$$[1] $\leftrightarrow$ eqn [5]

$$D(g) = \sum_{a,j,x} |a, j, x\rangle \langle a, j, x| D(g) \sum_{b,k,y} |b, k, y\rangle \langle b, k, y| \quad \text{(Completeness)}$$

$$= \sum_{a,j,x,b,k,y} |a, j, x\rangle\, \delta_{ab}\delta_{xy}[D(g)]_{jk}\, \langle b, k, y|$$

$$= \sum_{a,j,k,x} |a, j, x\rangle\, [D(g)]_{jk}\, \langle a, k, x|$$

This is another way of writing eqn [5]

Consequences of Schur's Lemma

Consider a commuting observable H :

$$[H, D(g)] = 0, \quad \forall g \in G$$

How much can symmetry alone tell us, without knowing the detailed physics of the system?

We want to constrain $\langle a, j, x| H |b, k, y\rangle$, first note that

$$P_{ax} = \sum_{j'} |a, j', x\rangle \langle a, j', x|$$

is the projection onto the invariant subspace labeled by a and x

Consequences of Schur's Lemma

Now since $P_{by}D(g)\,|b,k,y\rangle = D(g)\,|b,k,y\rangle$,

$$\begin{aligned} 0 &= \langle a,j,x|\,[H,D(g)]\,|b,k,y\rangle \\ &= \sum_{k'} \langle a,j,x|\,H\,\big|b,k',y\big\rangle\,\big\langle b,k',y\big|\,D(g)\,|b,k,y\rangle \\ &\quad - \sum_{j'} \langle a,j,x|\,D(g)\big|a,j',x\big\rangle\,\big\langle a,j',x\big|\,H\,|b,k,y\rangle \end{aligned}$$

Since we know $D(g)$'s matrix form, this simplifies to

$$\sum_{k'} \langle a,j,x|\,H\,\big|b,k',y\big\rangle\,[D_b(g)]_{k'k} = \sum_{j'} [D_a(g)]_{jj'}\,\big\langle a,j',x\big|\,H\,|b,k,y\rangle$$

Consequences of Schur's Lemma

Equivalently,

$$[H^{ab} D_b(g)]_{jk} = [D_a(g) H^{ab}]_{jk}$$

By **Schur's lemma ...**

1. $\langle a, j, x| \, H \, |b, k, y\rangle \propto \delta_{ab}$, since if $a \neq b$ H^{ab} vanishes
2. $\langle a, j, x| \, H \, |a, k, y\rangle \propto \delta_{jk}$, since H must be proportional to the identity

Conclusion :

$$\langle a, j, x| \, H \, |b, k, y\rangle = \delta_{ab}\delta_{jk} f_a(x, y) \tag{6}$$

Representation-Theoretic Origin of Degeneracy

All remaining physical dependence is confined in $f_a(x, y)$. An operator commuting with the symmetry cannot distinguish states within an irrep.

Outline

Orthogonality Relations

Consider the following dyadic linear operator:

$$A_{jl}^{ab} \equiv \sum_{g \in G} D_a(g^{-1}) \left| a, j \right\rangle \left\langle b, l \right| D_b(g)$$

where D_a and D_b are finite-dim irreps of G

$$\begin{aligned} D_a(g_1) A_{jl}^{ab} &= \sum_{g \in G} D_a(g_1 g^{-1}) \left| a, j \right\rangle \left\langle b, l \right| D_b(g) \\ &= \sum_{g' = g g_1^{-1}} D_a(g'^{-1}) \left| a, j \right\rangle \left\langle b, l \right| D_b(g'^{-1}) \\ &= A_{jl}^{ab} D_b(g_1) \end{aligned}$$

Orthogonality Relations

$D_a(g_1)A^{ab}_{jl} = A^{ab}_{jl}D_b(g_1)$ and by **Schur's lemma**, we conclude

$$A^{ab}_{jl} = \delta^{ab}\lambda^a_{jl}I$$

To determine λ^a_{jl}, we compute the trace when $a = b$ by two different ways

$$\begin{aligned}
\mathrm{tr}\left[A^{ab}_{jl}\right] &= \delta_{ab}\,\mathrm{tr}\left[\lambda^a_{jl}I\right] = \delta^{ab}\lambda^a_{jl}n_a \\
&= \delta^{ab}\sum_k \sum_{g\in G} \langle a, k| D_a(g^{-1})\,|a, j\rangle\,\langle b, l|\, D_b(g)|a, k\rangle \\
&= \delta^{ab}\sum_{g\in G} \langle a, l|\, D_a(g)D_a(g^{-1})\,|a, j\rangle = |G|\delta_{ab}\delta_{jl} \qquad \because \text{Completeness}
\end{aligned}$$

where n_a is the dimension of irrep D_a, and $|G|$ is order of the group

Orthogonality Relations

Therefore we conclude that

$$\lambda^{a}_{jl} = \frac{|G|}{n_a}\delta_{jl}, \quad A^{ab}_{jl} = \frac{|G|}{n_a}\delta_{ab}\delta_{jl}$$

Schur Orthogonality

$$\frac{1}{|G|}\sum_{g\in G} n_a [D_a(g^{-1})]_{kj}[D_b(g)]_{lm} = \delta_{ab}\delta_{jl}\delta_{km} \tag{7}$$

Especially for unitary irreps,

$$\frac{1}{|G|}\sum_{g\in G} n_a [D_a(g)]^{*}_{jk}[D_b(g)]_{lm} = \delta_{ab}\delta_{jl}\delta_{km} \tag{8}$$

Orthogonality Relations

Equivalently,

$$\sum_{g\in G}\left[\sqrt{\frac{n_a}{|G|}}D_a(g)\right]^*_{jk}\left[\sqrt{\frac{n_b}{|G|}}D_b(g)\right]_{lm}=\delta_{ab}\delta_{jl}\delta_{km} \tag{9}$$

So with proper normalization, the matrix elements of the inequivalent unitary irreps

$$\left[\sqrt{\frac{n_a}{|G|}}D_a(g)\right]_{jk}$$

are **orthonormal** functions of g, furthermore

Theorem

The matrix elements of the unitary irreps of G are a **complete orthonormal set** for the vector space of regular rep, or alternatively, for **function space** of $g \in G$.

Orthogonality Relations

An arbitrary function $F(g) : G \to \mathbb{C}$ can be written in terms of a **bra vector** in the regular rep space :

$$F(g) = \langle F|g\rangle = \langle F|\, D_R(g)\, |e\rangle$$

($\langle F|$ is a **linear functional** (dual vector) of vectors $|g\rangle$ in regular vector space)

$$\langle F| = \sum_{g' \in G} \langle F|g'\rangle \langle g'| \tag{10}$$

$$F(g) = \langle F|g\rangle = \sum_{g' \in G} \langle F|g'\rangle \langle g'|\, D_R(g)\, |e\rangle \tag{11}$$

$$= \sum_{g' \in G} \langle F|g'\rangle \left[D_R(g)\right]_{g' e} \tag{12}$$

Orthogonality Relations - Example

Since D_R is completely reducible, we can express $F(g)$ as a **linear combination of the matrix elements of the irreps**!

Cor. Let $|G|$ be the order of the group and n_i be the dimension of the irrep D_i. Then we have

$$|G| = \sum_i n_i^2.$$

- **Discrete Fourier Series**
  A cyclic group $\mathbb{Z}/N\mathbb{Z}$, with irreps $D_n(a^j) = \exp[(2\pi i)n(j/N)]$
  Schur orthogonality reads

$$\frac{1}{N}\sum_{j=1} e^{-2\pi i n' j/N} e^{2\pi i m' j/N} = \delta_{nn'}$$

Generalization to Compact Continuous Groups

Angular momentum state $|L, M\rangle$ transforming under rotation (L integer):

$$\mathbf{R}\,|L, M\rangle = \sum_{M'} D^{(L)}_{MM'}\,|L, M'\rangle$$

Wigner D-matrices $D^{(L)}_{MM'}$ form unitary irreps of continuous group $SO(3)$
(We use *Euler angle* parametrization $D^{(l)}_{mm'}(\alpha, \beta, \gamma)$)

Generalization :

$$\frac{1}{|G|}\sum_{g\in G} \to \frac{1}{\mathrm{vol}(G)}\int d\mu(g)$$

Or we may use some invariant normalized measure $d\mu$ by taking $\mathrm{vol}(G) = 1$, known as the **Haar measure**

Generalization to Compact Continuous Groups

We can find 'volume' of $SO(3)$, corresponding to $|G|$

$$\mathrm{vol}(G) = \int_0^{2\pi} d\alpha \int_0^{\pi} \sin\beta \, d\beta \int_0^{2\pi} d\gamma = 8\pi^2$$

Schur orthogonality :

$$\int_{R\in SO(3)} d\mathbf{R} \, D^{(l)}_{mn} D^{(l')*}_{m'n'} = \frac{8\pi^2}{2l+1} \delta_{ll'} \delta_{mm'} \delta_{nn'}$$

Using the fact that

$$D^{(l)}_{m0}(\phi, \theta, 0) = \sqrt{\frac{4\pi}{2l+1}} Y_l^{m*}(\theta, \phi)$$

Generalization to Compact Continuous Groups

Integrating out 2π redundancy,

$$\int_{S^2} d\Omega D^{(l)}_{m0} D^{(l')*}_{m'0} = \frac{4\pi}{\sqrt{2l+1}\sqrt{2l'+1}} \int_{S^2} d\Omega\, Y^{m*}_{l}\, Y^{m'}_{l'}$$
$$= \frac{4\pi}{2l+1}\delta_{ll'}\delta_{mm'}$$

We have **orthogonality of spherical harmonics**

$$\int_{S^2} d\Omega\, Y^{m*}_{l}\, Y^{m'}_{l'} = \delta_{ll'}\delta_{mm'} \tag{13}$$

Any L^2-function on sphere $\psi : S^2 \to \mathbb{C}$ can be expanded in spherical harmonics!

Outline

Proof of The Useful Theorem

Useful Theorem

Every representation of a finite group is equivalent to a unitary representation.

proof. Let $D(g)$ be a representation of a finite group G. Define the operator

$$S = \sum_{g \in G} D(g)^\dagger D(g).$$

Since S is hermitian and positive-semidefinite, S can be diagonalized.

$$S = U^{-1} d U, \quad d = \begin{pmatrix} d_1 & 0 & \cdots \\ 0 & d_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \quad d_i \geq 0$$

Proof of The Useful Theorem

Proposition. $d_i > 0, \quad \forall i$.

proof. We show that $d_i > 0 \quad \forall i$. Suppose that there exists some zero d_λ. Then there exists a nonzero vector λ such that

$$\lambda^\dagger S \lambda = 0 = \sum_{g \in G} \|D(g)\lambda\|^2$$

This implies $D(g)\lambda$ are zero for all $g \in G$. Contradiction, since $D(e) = 1$.

Since all $d_i > 0$, we can define 'square root' of S :

$$\sqrt{S} \equiv U^{-1} \begin{pmatrix} \sqrt{d_1} & 0 & \cdots \\ 0 & \sqrt{d_2} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} U$$

Proof of The Useful Theorem

Clearly $\det \sqrt{S} \neq 0$, so $\sqrt{S}$ is invertible. Now define

$$D'(g) = \sqrt{S} D(g) \sqrt{S}^{-1}.$$

To conclude the proof, we verify that $D'(g)$ is a unitary rep.

$$\begin{aligned} D'(g)^\dagger D'(g) &= \sqrt{S}^{-1} D(g)^\dagger S D(g) \sqrt{S}^{-1} \\ &= \sqrt{S}^{-1} \sum_{h \in G} D(g)^\dagger D(h)^\dagger D(h) D(g) \sqrt{S}^{-1} \\ &= \sqrt{S}^{-1} \sum_{h \in G} D(hg)^\dagger D(hg) \sqrt{S}^{-1} \\ &= 1. \quad \square \end{aligned}$$

Direct Sum

Direct Sum

Let $W_1, \cdots, W_n$ be subspaces of a vector space V. V is a **direct sum** of $W_1, \cdots, W_n$ if for every element v of V there exist *unique* elements $w_i \in W_i$ such that

$$v = w_1 + \cdots w_n.$$

Complementary Subspace

Let V be a finite-dimensional vector space. Let $W \leq V$. Then there exists a subspace (complement) U such that

$$V = W \oplus U.$$

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